

SHREYAS RAORANE

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Education

University of Pennsylvania – *Master of Science in Engineering (MSE), Robotics, GPA: 3.8/4.0*

Aug 2024 – May 2026

Work Experience

Time Critical Software Engineering Intern – *OPEX Corporation (warehouse robotics)*

Jun 2025 – Aug 2025

- Engineered a multi-threaded discrete-event hardware-in-the-loop (HIL) simulation in C++ for warehouse automation, managing 1,000+ parallel material flow events with shared state across 3 concurrent threads
- Developed a time-critical scheduling engine completing actuation sequences within 3 ms, modeling real hardware timing so the software team could run automated regression and validation tests against the simulator instead of physical hardware

Software Development Engineer – *Brillio Technologies (Verizon ESAP enterprise platform)*

Jun 2022 – Jun 2024

- Sole developer on the Wholesale EMEA module of a Verizon enterprise platform serving 10,000+ users: owned the UI end-to-end and optimized PostgreSQL queries to cut response times by 35%
- Delivered features through the full software development life cycle: code reviews, source control (Git), CI/CD, and production support

Robotics Projects

MASTER'S THESIS: MULTI-ROBOT COORDINATION UNDER DEGRADED COMMUNICATION

Oct 2025 – May 2026

GRASP Lab – Advisors: Prof. Phan, Prof. Mangharam, Prof. Hsieh

[\[Code\]](#)

- Designed a decentralized task-allocation algorithm (LCBA) that keeps warehouse robot fleets completing pick-and-deliver tasks even when the communication network splits into disconnected groups – settings where both centralized and state-of-the-art decentralized baselines drop to zero task completion
- Evaluated across 432+ runs spanning six warehouse environments (6–200 robots) in a full factorial sweep (6 communication ranges, 12 task-arrival rates, 3 path planners, 4 allocation methods): LCBA held flat throughput across the connectivity spectrum while every baseline degraded sharply below the connectivity threshold
- Outperformed the centralized greedy allocator on fully connected graphs (0.91 makespan ratio) via insertion-based bidding and event-driven replanning; built a planner-agnostic allocation layer integrating three interchangeable multi-agent path planners (Cooperative A*, PBS, RHCR)

SBAMP: SAMPLING-BASED ADAPTIVE MOTION PLANNING – ICRA 2026 WORKSHOP PAPER

Mar 2025 – Apr 2025

University of Pennsylvania

[\[Code\]](#) [\[Paper\]](#)

- First-author paper published at the ICRA 2026 Workshop on Frontiers of Optimization for Robotics**, addressing the trade-off between computationally expensive global planning and fast local adaptation in dynamic environments
- Coupled RRT* global planning with an online Lyapunov-stable dynamical-systems controller requiring no pre-trained data, embedding lightweight constrained optimization in the control loop for stable real-time adaptation that preserves global path structure
- Validated in simulation and on RoboRacer (F1/10) hardware: robust recovery from physical disturbances, reliable obstacle handling, and moving-goal tracking where static RRT* replanning could not keep up

AUTONOMOUS RACING STACK (RoboRacer / F1/10 Platform, ROS 2)

Feb 2025 – May 2025

University of Pennsylvania

[\[Code\]](#)

- Built and raced the full autonomy stack on a 1/10-scale autonomous race car (LIDAR + IMU): SLAM mapping, particle-filter localization, global raceline optimization, pure-pursuit tracking with a reactive gap-follow overtaking layer, and PID wall-following – reached the semifinals of Penn's head-to-head race and time-qualified for the ICRA 2025 RoboRacer Grand Prix
- Developed the supporting toolchain: LIDAR time-to-collision emergency braking, a custom ROS 2 waypoint mapping/logging pipeline with RViz visualization, and real-time telemetry + interactive parameter tuning (Foxglove, Gradio UI)
- Bridged AutoDRIVE Simulator and F1/10 hardware over a shared ROS 2 interface, tuning parameters across the sim-to-real gap; added an RGB-D camera feed for scene visibility alongside LIDAR

SwarmSync: DISTRIBUTED MPC FOR QUADROTOR SWARMS

Oct 2025 – Dec 2025

University of Pennsylvania

[\[Code\]](#)

- Extended an online distributed model-predictive control (DMPC) framework (Luis et al.) with moving goals and an online Hungarian task-reallocation layer that re-assigns drones to goals to minimize total travel distance
- Generated collision-free trajectories via per-agent QP trajectory optimization; scaled to 32-drone demos and a 64-agent sweep with zero collisions and 100% goal success across all 9 scenarios at >200 Hz solve rate

Skills

Programming: C++, Python, SQL, MATLAB, CUDA, React, Object-Oriented Design, Design Patterns

Machine Learning: PyTorch, TensorFlow/Keras, JAX, Deep Learning, Anomaly Detection, LLM Integration (OpenAI API)

Systems: Multi-threading, Real-time Systems, Low-Latency, Memory Management, Distributed Systems, Event-driven Architecture, Linux

Perception & Localization: Computer Vision, Sensor Fusion (EKF, Quaternion UKF, Particle Filter), SLAM (slam_toolbox), Localization & Mapping (robot_localization), Object Detection (YOLO), Sensor Integration (LIDAR, Cameras, IMU, RGB-D, Odometry)

Motion Planning & Control: Path Planning, RRT*, A*, Pure Pursuit, Model Predictive Control (MPC), DMPC, MPPI, iLQR, Trajectory Optimization (QP), Minimum-Snap Trajectories, Raceline Optimization, Sampling-Based Planning, Collision Avoidance

Robotics & Tools: ROS, ROS 2 (ROS2), Nav2 (custom costmap plugins), TurtleBot3, NVIDIA Jetson, Gazebo, IsaacSim, MuJoCo, PyBullet, Foxglove, RViz, HIL, Docker, CMake, Git, CI/CD